

E-commerce → Order → Status, Ratings, Delivery Date
→ Inventory →

Banking System → Transaction - Account Balance
→ Update personal details

Database Operations CRUD

C	→ Create	→ CREATE	✓
R	→ Read	→ SELECT	✓
U	→ Update	→ UPDATE	←
D	→ Delete	→ DELETE	

Update

The ****UPDATE operation**** is used to modify the values of one or more attributes (columns) for selected tuples (rows) in a relation.

2 step process

- 1) Selection
- 2) Modification

To ensure correctness, UPDATE must be

- 1) Precise
- 2) Validation
- 3) Safe
- 4) Consistent

Best practices

- 1) Preview
- 2) Verify
- 3) Understand the impact

Syntax

UPDATE table_name

SET column1 = value1, column2 = value2, ...

WHERE condition;

Q1 Update credits for CS101 to 4

Q2 for student 1 → failed course 2
Re-enroll.

Update enrollment date & grade to NULL
to today

SET operations support
↳ Arithmetic

Point list of all course name, credit, instructor name for all courses.

JOINS

A ****JOIN**** is an operation that combines tuples from two relations based on a condition.

courses				instructors	
name	credits	i_id		i_id	name
AI	2	1		1	Ayush
HLID	4	2		2	Kshiti
LLID	4	2			

Left Right

▷ Cartesian product (x)

name	credits	i_id	i_id	name
AI	2	1	1	Ayush
AI	2	1	2	Kshiti
HLID	4	2	1	Ayush

LLD	4	2	2	Kshity
LLD	4	2	1	Agush
LLD	4	2	2	Kshity

2) Selection (σ)

Type of Joins

- 1) Inner Join
 - 2) Left Join
 - 3) Right Join
 - 4) Full Outer Join (Not in MySQL)
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Types of Relations (Cardinality)

- 1) One to One
- 2) One to Many / Many to One
- 3) Many to Many

Eg 1 Passport Database

Users

user-id	name	dob	
1	Ayush	.	
2	Sonali	.	
3	Zahara	.	

Passport

passport-id	pass-no	expiry-date	user-id
1	P101	"	1
2	P102	"	2
3	P103	"	3

for all enrollments, print

<u>course_name</u>	<u>student_name</u>	<u>instructor_name</u>	credits	<u>grade</u>
↓	↓	↓	↓	↓
course	student	instructor	cores	course-endl.